

Categorical Time-Series Analysis

General Notation

Let (X_t) be a *categorical* time series process, where each X_t takes values in

$$S := \{s_0, \dots, s_m\}, \quad m \in \mathbb{N}.$$

(X_t) may be *nominal*, in which case the order of magnitude of the elements in S is arbitrary and solely used for notational consistency, or *ordinal*, in which case the order of the elements in S reflects the inherent order of the elements.

Furthermore, (X_t) is assumed to be stationary. A distinction is made between two forms of stationarity. A process is said to be weakly stationary if:

$$\begin{aligned} E(X_t) &= \mu_X & \forall t, \\ E(X_t^2) &< \infty & \forall t, \\ \text{Cov}(X_t, X_{t+h}) &= \gamma(h) & \forall t, h. \end{aligned}$$

In contrast, strong stationarity requires that all finite-dimensional distributions are invariant under time shifts, i.e.,

$$F_{X_t, \dots, X_{t+j}} = F_{X_{t+h}, \dots, X_{t+h+j}} \quad \forall t, j, h$$

Marginal Distributions

Regardless of whether we consider a simple categorical process with few categories or attempt to describe a complex, temporally evolving process using moments and measures of serial dependence, the analysis of categorical processes is always built on examination of the marginal probabilities. However, because ordinal and nominal data encode different types of information, their marginals are typically represented in different ways:

- **Ordinal data:** Analytical tools for ordinal processes must preserve the natural ordering of the elements in S . Measures that ignore this ordering—such as the mode—would discard essential information about the relative positions of the categories. For this reason, methods for ordinal data rely on the marginal cumulative distribution function (CDF), which accumulates probabilities in accordance with the natural ordering of the categories, thereby preserving the ordinal structure, and is therefore the standard representation for ordinal data:

$$\mathbf{f} = (f_0, \dots, f_{m-1})^\top, \quad f_i := P(X_t \leq s_i).$$

Although one could equivalently estimate the PMF and derive the CDF from it, representing ordinal data through the CDF is often more convenient and aligns with widely used modelling frameworks.

- **Nominal data:** Since the categories of nominal variables do not possess a meaningful order, statistical procedures cannot make use of cumulative structure. Therefore the marginal distribution is best represented by the probability mass function (PMF):

$$\mathbf{p} = (p_0, \dots, p_{m-1})^\top, \quad p_i := P(X_t = s_i).$$

Bivariate Probabilities at Lag h

Temporal dependence can be described by bivariate (lag- h) probabilities:

- **Ordinal series (cumulative):**

$$f_{ij}(h) := P(X_t \leq s_i, X_{t+h} \leq s_j), \quad i, j < m.$$

- **Nominal series (joint):**

$$p_{ij}(h) := P(X_t = s_i, X_{t+h} = s_j), \quad i, j \leq m.$$

These quantities serve as input for dependence measures such as association coefficients or transition structures.

Location

For ordinal data, the measure of location is given by the median and for nominal its the mode

$$\begin{aligned} Med(X_{t \text{ (ord.)}}) &= \min(s_i \in S : F(s_i) \geq \frac{1}{2}) \\ Mod(X_{t \text{ (nom.)}}) &= s_i \in S : P(X_t = s_i) = \operatorname{argmax}(\mathbf{p}) \end{aligned}$$

Measures of Dispersion

Dispersion of categorical variables is commonly quantified via the *Index of Ordinal Variation* (IOV) for ordinal data and the *Index of Qualitative Variation* (IQV) for nominal data. Both measures take values in the interval $[0, 1]$, where 0 indicates no dispersion (degenerate one-point distribution) and 1 indicates maximal dispersion.

For ordinal variables, dispersion is defined in terms of the CDF vector $\mathbf{f}$:

$$\text{IOV} = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i), \quad \text{where } f_i = P(X_t \leq s_i)$$

For nominal variables, dispersion is defined in terms of the PMF $\mathbf{p}$:

$$\text{IQV} = \frac{m+1}{m} \left(1 - \sum_{i=0}^m p_i^2 \right), \quad p_i = P(X_t = s_i).$$

Minimal Dispersion

Both measures attain the value 0 if and only if the distribution is degenerate, that is, if all probability mass is concentrated in a single category.

Ordinal case. A one-point distribution is characterized by a CDF vector $\mathbf{f}$ of the form

$$\mathbf{f} \in \left\{ \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 1 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}.$$

In each case, at every index i either $f_i = 0$ or $f_i = 1$, hence

$$f_i(1 - f_i) = 0 \quad \forall i.$$

Therefore,

$$\text{IOV} = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i) = 0.$$

Nominal case. A degenerate distribution has a PMF of the form

$$\mathbf{p} \in \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}.$$

Thus,

$$\sum_{i=0}^m p_i^2 = m \cdot 0 + 1 \cdot 1 = 1,$$

and therefore

$$\text{IQV} = \frac{m+1}{m}(1 - 1) = 0.$$

Maximal Dispersion

Ordinal case. Dispersion of an ordinal variable is maximal when half of the probability mass lies in the lowest category and half in the highest category; all intermediate categories receive probability zero. This results in a CDF vector

$$\mathbf{f} = \left(\frac{1}{2}, \frac{1}{2}, \dots, \frac{1}{2} \right)^\top.$$

Then

$$f_i(1 - f_i) = \frac{1}{2} \left(1 - \frac{1}{2} \right) = \frac{1}{4} \quad \forall i,$$

and thus

$$\text{IOV} = \frac{4}{m} \cdot \sum_{i=0}^{m-1} \frac{1}{4} = \frac{4}{m} \cdot \frac{m}{4} = 1.$$

Nominal case. Maximal dispersion is obtained when probability mass is distributed uniformly across all $m + 1$ categories:

$$p_i = \frac{1}{m+1}, \quad i = 0, \dots, m.$$

Then

$$\sum_{i=0}^m p_i^2 = (m+1) \frac{1}{(m+1)^2} = \frac{1}{m+1},$$

so that

$$\text{IQV} = \frac{m+1}{m} \left(1 - \frac{1}{m+1} \right) = \frac{m+1}{m} \cdot \frac{m}{m+1} = 1.$$

Samples Measures

For the estimation of the (cumulative) probabilities based on samples we use the (cumulative) relative frequencies, that can be best expressed if switching from (X_t) to its binarization ($(\mathbf{Z}_t)$ for ordinal variables, $(\mathbf{Y}_t)$ for nominal):

$$\mathbf{Z}_t = (Z_{t,0}, \dots, Z_{t,m-1})^T, Z_{t,i} = I_{\{X_t \leq s_i\}} \quad \mathbf{Y}_t = (Y_{t,0}, \dots, Y_{t,m})^T, Y_{t,i} = I_{\{X_t = s_i\}}$$

Now we can estimate $\mathbf{p}/\mathbf{f}$ by taking the mean about the binarization:

$$\hat{\mathbf{f}} = \frac{1}{n} \sum_{t=1}^n \mathbf{Z}_t, \quad \hat{\mathbf{p}} = \frac{1}{n} \sum_{t=1}^n \mathbf{Y}_t$$

To get the asymptotics of the different sample measures, we first need a CLT on $\hat{\mathbf{f}}/\hat{\mathbf{p}}$. As our data are not (necessarily) independent, the CLT only holds for a stationary process that fulfills a strong mixing condition (e.g. alpha mixing; Bradley 2005). The stationarity of $\mathbf{Z}_t/\mathbf{Y}_t$ follows from the stationarity of X_t , as the transforming function is deterministic.

A sequence is said to be α -mixing, if:

For any event $A \in \sigma(X_s : s \leq t)$ and event $B \in \sigma(X_s : s \geq t+h)$ the α -coefficient

$$\alpha(h) = \sup |P(A \cup B) - P(A)P(B)| \text{ goes to } 0 \text{ for } h \rightarrow \infty$$

If α -mixing is satisfied it holds that:

$$\sqrt{n}(\overbrace{\bar{\mathbf{Z}}_t}^{\hat{\mathbf{f}}} - \mathbf{f}) \xrightarrow{d} \mathcal{N}(0, \mathbf{\Sigma}), \quad \text{with } \mathbf{\Sigma} = (\sigma_{i,j})_{i,j=0, \dots, m-1}$$

$$\text{where } \sigma_{i,j} = \text{Cov}(Z_{0,i}, Z_{0,j}) + \sum_{h=1}^{\infty} (\text{Cov}(Z_{0,i}, Z_{h,j}) + \text{Cov}(Z_{h,i}, Z_{0,j}))$$

$$\text{and } \text{Cov}(Z_{t,i}, Z_{s,j}) = \begin{cases} f_{ij}(t-s) - f_i f_j & \text{if } t \geq s \\ f_{ji}(s-t) - f_i f_j & \text{if } s \geq t \end{cases}$$

To now derive the asymptotics of the dispersion measures (scalar functions) we can apply the delta method (i.e. Theorem on 5.25 on Asymptotic Distribution of $g(\mathbf{X}_n)$ in the formulary) according to which the following holds:

$$n^{\frac{1}{2}}(IOV(\hat{\mathbf{f}}) - IOV(\mathbf{f})) \xrightarrow{d} \mathcal{N}(0, \mathbf{G}\Sigma\mathbf{G})$$

$$\text{Where } G_{1 \times m} = \left[\frac{\partial IOV(\mathbf{f})}{\partial \hat{f}_0}, \dots, \frac{\partial IOV(\mathbf{f})}{\partial \hat{f}_{m-1}} \right]'$$

The derivations for other dispersion metrics are analogous.

Modelling Missingness

To model the missingness amplitude modulation is used. The concept stems from physics and allows to model a stochastic process that runs concurrent to another one and modulates its amplitude. In the case of missingness the main process $(X_t)_t$ is either there or not there. Therefore we can define our amplitude modulating process as

$$O_t \in \{0, 1\} \quad \forall t$$

where $O_t = 1$ if (X_t) is observed and 0 if it isn't